

# **1. Project Title**

LabTrack – University Lab Equipment Management System

## **1.1 Project Overview**

A web-based lab equipment management system designed for universities where students can search, request, borrow, and return lab equipment through an online platform, while teachers and lab administrators can approve requests, monitor equipment usage, manage inventory, and generate reports. The system focuses on efficient database management, automation, data consistency, and secure transaction handling using Oracle Database Express Edition with PL/SQL and a website interface.

## **1.2 Project Objective**

To develop a smart, database-driven university laboratory equipment management system that automates equipment borrowing, inventory tracking, approval workflows, fine management, and report generation while ensuring data integrity and efficient resource utilization.

# **2. Key Features**

## **2.1. User Authentication & Role Management**

- Student login/register
- Teacher/Supervisor login
- Lab Assistant/Admin panel access
- Role-based permissions (Student / Teacher / Lab Assistant)

## **2.2. Equipment Inventory Management**

- Add, update, delete equipment
- Equipment category management
- Lab-wise equipment organization
- Stock availability tracking
- Equipment status (Available / Borrowed / Maintenance)

## **2.3. Smart Search, Sort & Filter**

- Search equipment by name
- Filter by category, lab, availability
- Sort by popularity, availability, latest added

## **2.4. Equipment Borrowing System**

- Online equipment request submission
- Borrow scheduling (borrow & return date)
- Quantity selection
- Approval/rejection workflow

## **2.5. Approval Management**

- Teacher/Admin approval system
- Request tracking status:
  - Pending
  - Approved
  - Rejected
  - Returned

## **2.6. Borrowing History & Tracking**

- Student borrowing history
- Current borrowed items tracking
- Return history records

## **2.7. Fine & Damage Management**

- Late return fine calculation
- Damage report submission
- Penalty tracking

## **2.8. Automatic Stock Management**

- Stock decreases automatically after approval
- Stock increases automatically after return
- Inventory consistency maintenance

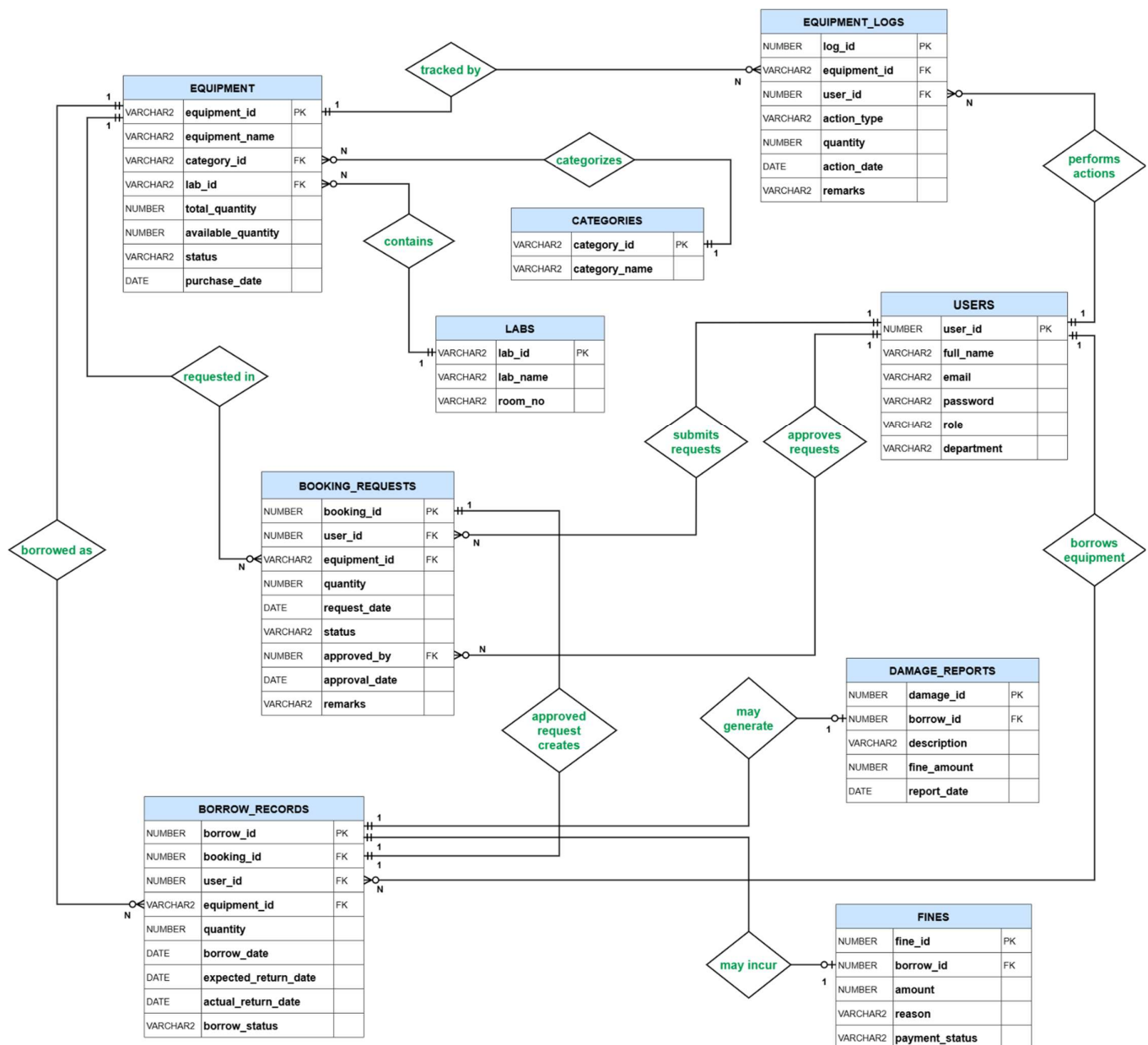
## **2.9. Reports & Analytics Dashboard**

- Most borrowed equipment
- Equipment usage statistics
- Lab-wise borrowing reports
- Monthly borrowing analysis

## 2.10. Database & PL/SQL Features

- DDL operations (CREATE, ALTER, DROP)
- DML operations (INSERT, UPDATE, DELETE, SELECT)
- JOIN, GROUP BY, HAVING, Subqueries
- Transaction Management (COMMIT / ROLLBACK)
- Constraints & Domain Integrity
- PL/SQL Procedures, Functions, Triggers
- Automated database actions using triggers

## 3. Entity Relationship Diagram



### 3.1 Relationship Explanation:

| Relation                                         | Cardinality | Relationship Description                                                                                                                  |
|--------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Users → Booking Requests                         | 1 : Many    | One user can submit multiple booking requests, but each booking request belongs to exactly one user.                                      |
| Users (Teacher/Lab Assistant) → Booking Requests | 1 : Many    | One teacher/lab assistant can approve multiple booking requests, but each approved request is approved by only one teacher/lab assistant. |
| Categories → Equipment                           | 1 : Many    | One category can contain multiple equipment items, but each equipment item belongs to only one category.                                  |
| Labs → Equipment                                 | 1 : Many    | One lab can contain multiple equipment items, but each equipment item is assigned to only one lab.                                        |
| Equipment → Booking Requests                     | 1 : Many    | One equipment item can be requested many times by different users, but each booking request is for one specific equipment item.           |
| Booking Requests → Borrow Records                | 1 : 1       | An approved booking request creates exactly one borrow record, and each borrow record originates from exactly one booking request.        |
| Users → Borrow Records                           | 1 : Many    | One user can borrow equipment multiple times, but each borrow record belongs to only one user.                                            |
| Equipment → Borrow Records                       | 1 : Many    | One equipment item can appear in many borrowing records over time, but each borrow record references one equipment item.                  |
| Borrow Records → Damage Reports                  | 1 : 0..1    | A borrow record may generate zero or one damage report depending on whether damage occurred.                                              |
| Borrow Records → Fines                           | 1 : 0..1    | A borrow record may generate zero or one fine depending on overdue returns or penalties.                                                  |
| Equipment → Equipment Logs                       | 1 : Many    | One equipment item can have many log entries tracking stock changes, borrowing, returns, and damage events.                               |
| Users → Equipment Logs                           | 1 : Many    | One user can perform multiple actions recorded in logs, while each log entry is associated with one user.                                 |

**3.2 Work Flow:** A student submits a booking request. Once approved by a teacher or lab assistant, exactly one borrows record is created. A borrow record may optionally generate a fine and/or damage report. All inventory-related activities are tracked in equipment logs.

## **4 Database Schema**

The system is built on 10 core relational tables:

### **4.1. USERS**

Stores information about all system users including students, teachers, and lab assistants.

| Attribute    | Data Type     | Description                                 |
|--------------|---------------|---------------------------------------------|
| user_id (PK) | NUMBER(10)    | Unique user identifier                      |
| full_name    | VARCHAR2(100) | User's full name                            |
| email        | VARCHAR2(100) | User email address                          |
| password     | VARCHAR2(255) | User password                               |
| role         | VARCHAR2(20)  | User role (Student, Teacher, Lab Assistant) |
| department   | VARCHAR2(50)  | Department name                             |

### **4.2. LABS**

Stores information about university laboratories.

| Attribute   | Data Type     | Description                  |
|-------------|---------------|------------------------------|
| lab_id (PK) | VARCHAR2(10)  | Unique laboratory identifier |
| lab_name    | VARCHAR2(100) | Laboratory name              |
| room_no     | VARCHAR2(20)  | Laboratory room number       |

### **4.3. CATEGORIES**

Stores equipment categories.

| Attribute        | Data Type     | Description                |
|------------------|---------------|----------------------------|
| category_id (PK) | VARCHAR2(10)  | Unique category identifier |
| category_name    | VARCHAR2(100) | Category name              |

#### 4.4. EQUIPMENT

Stores all laboratory equipment information.

| Attribute          | Data Type     | Description                 |
|--------------------|---------------|-----------------------------|
| equipment_id (PK)  | VARCHAR2(10)  | Unique equipment identifier |
| equipment_name     | VARCHAR2(150) | Equipment name              |
| category_id (FK)   | VARCHAR2(10)  | References Categories table |
| lab_id (FK)        | VARCHAR2(10)  | References Labs table       |
| total_quantity     | NUMBER(10)    | Total stock quantity        |
| available_quantity | NUMBER(10)    | Available quantity          |
| status             | VARCHAR2(30)  | Equipment status            |
| purchase_date      | DATE          | Purchase date               |

#### 4.5. BOOKING\_REQUESTS

Stores equipment borrowing requests submitted by users.

| Attribute         | Data Type     | Description                |
|-------------------|---------------|----------------------------|
| booking_id (PK)   | NUMBER(10)    | Unique booking identifier  |
| user_id (FK)      | NUMBER(10)    | References Users table     |
| equipment_id (FK) | VARCHAR2(10)  | References Equipment table |
| quantity          | NUMBER(10)    | Requested quantity         |
| request_date      | DATE          | Request submission date    |
| status            | VARCHAR2(20)  | Request status             |
| approved_by (FK)  | NUMBER(10)    | Approver user ID           |
| approval_date     | DATE          | Approval date              |
| remarks           | VARCHAR2(255) | Additional remarks         |

## 4.6. BORROW\_RECORDS

Stores information about approved borrowing transactions.

| Attribute            | Data Type    | Description                       |
|----------------------|--------------|-----------------------------------|
| borrow_id (PK)       | NUMBER(10)   | Unique borrowing identifier       |
| booking_id (FK)      | NUMBER(10)   | References Booking Requests table |
| user_id (FK)         | NUMBER(10)   | References Users table            |
| equipment_id (FK)    | VARCHAR2(10) | References Equipment table        |
| quantity             | NUMBER(10)   | Borrowed quantity                 |
| borrow_date          | DATE         | Borrow date                       |
| expected_return_date | DATE         | Expected return date              |
| actual_return_date   | DATE         | Actual return date                |
| borrow_status        | VARCHAR2(20) | Borrowing status                  |

## 4.7. DAMAGE\_REPORTS

Stores damaged equipment reports.

| Attribute      | Data Type     | Description                     |
|----------------|---------------|---------------------------------|
| damage_id (PK) | NUMBER(10)    | Unique damage report identifier |
| borrow_id (FK) | NUMBER(10)    | References Borrow Records table |
| description    | VARCHAR2(500) | Damage description              |
| fine_amount    | NUMBER(10,2)  | Damage penalty amount           |
| report_date    | DATE          | Report submission date          |

## 4.8. FINES

Stores fine information for overdue returns or damages.

| Attribute      | Data Type     | Description                     |
|----------------|---------------|---------------------------------|
| fine_id (PK)   | NUMBER(10)    | Unique fine identifier          |
| borrow_id (FK) | NUMBER(10)    | References Borrow Records table |
| amount         | NUMBER(10,2)  | Fine amount                     |
| reason         | VARCHAR2(255) | Fine reason                     |
| payment_status | VARCHAR2(20)  | Fine payment status             |

#### 4.9. EQUIPMENT\_LOGS

Stores equipment activity and inventory logs.

| Attribute         | Data Type     | Description                |
|-------------------|---------------|----------------------------|
| log_id (PK)       | NUMBER(10)    | Unique log identifier      |
| equipment_id (FK) | VARCHAR2(10)  | References Equipment table |
| user_id (FK)      | NUMBER(10)    | References Users table     |
| action_type       | VARCHAR2(30)  | Type of activity           |
| quantity          | NUMBER(10)    | Quantity affected          |
| action_date       | DATE          | Activity date              |
| remarks           | VARCHAR2(255) | Additional remarks         |

#### 5. Technology Stack

| Layer    | Technology                           |
|----------|--------------------------------------|
| Frontend | HTML5, CSS3, Bootstrap 5, JavaScript |
| Backend  | Laravel, PHP                         |
| Database | MySQL                                |
| Tools    | Composer, Git, GitHub                |
| Server   | Apache (XAMPP)                       |

#### Expected Outcome

A complete university lab management website that digitizes equipment borrowing and inventory processes while demonstrating practical implementation of database concepts, SQL queries, PL/SQL programming, automation, and web-based management.